

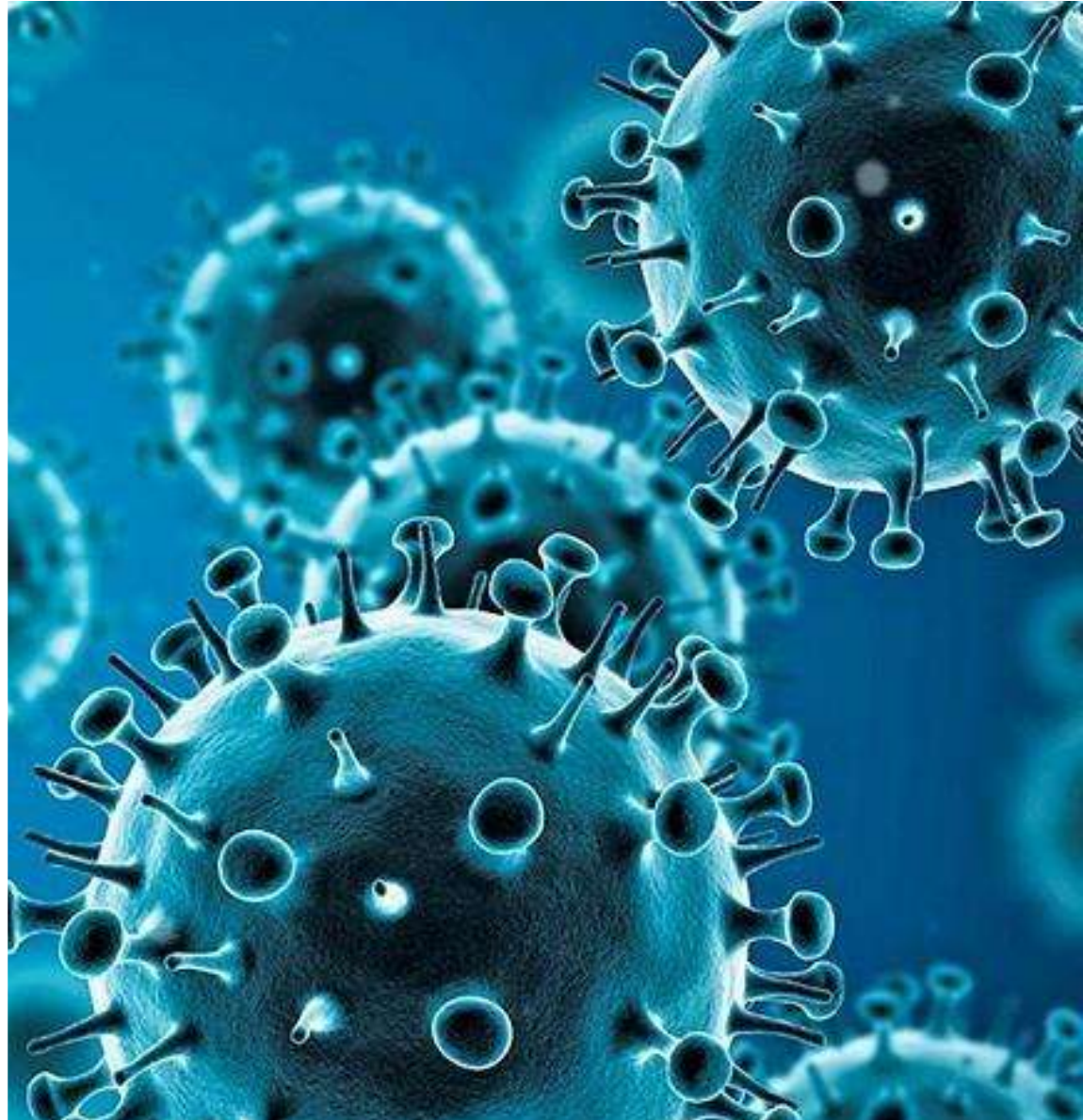
COVID-19 and vaccination

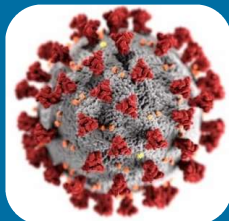
Video Link:

<https://vimeo.com/655499844/251ef55ae4>

Dr Mel Bradley

9th December 2021





**SARS-CoV-2 infection,
pathology and transmission**



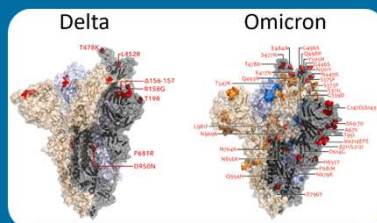
**The clinical trial process and
COVID-19 vaccine approval**



COVID-19 vaccines



**Child vaccination and booster
vaccinations**

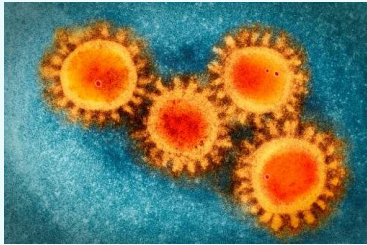


COVID-19 variants



COVID-19 and the future

SARS-CoV-2 (COVID-19)



Coronaviruses are a diverse group of retroviruses infecting many different animals



SARS, MERS and some types of 'common cold' are caused by different coronavirus



SARS-CoV-2 first reported in Wuhan, China in December 2019



May have emerged from a recombination of bat and pangolin coronaviruses

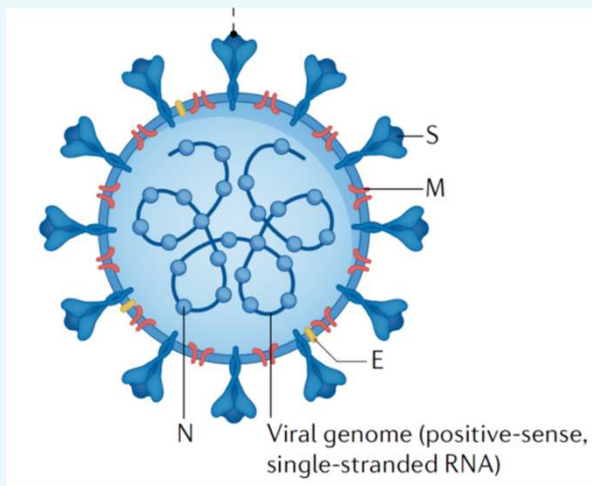


Declared a pandemic in March 2020 due to transmission to almost all countries of the world

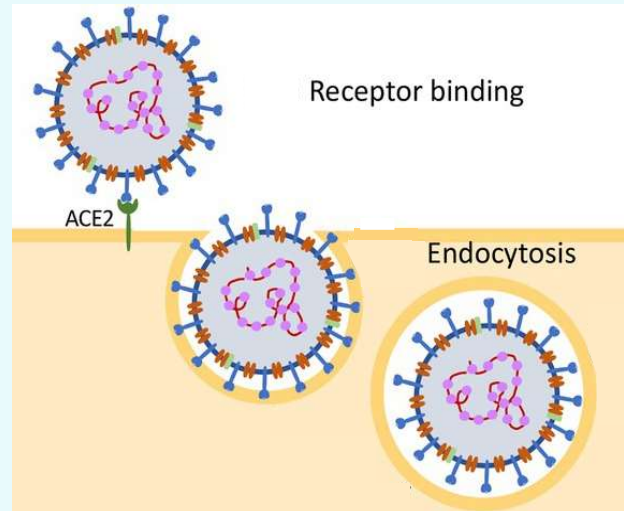


As of September 2021, 219 million people have been infected and 4.55 million people have died of it

SARS-CoV-2 (COVID-19)

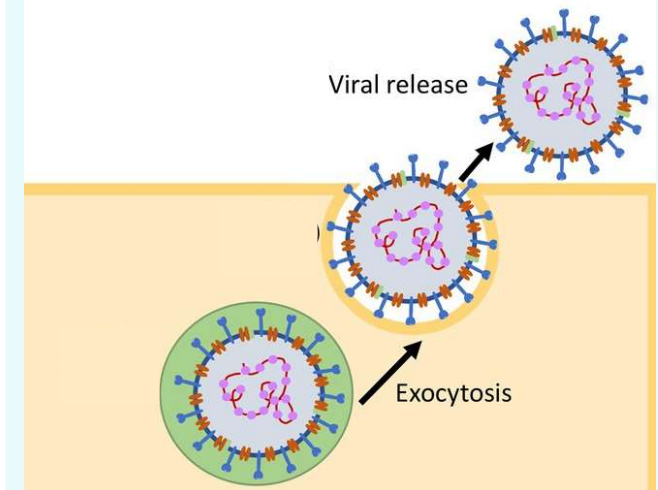


S: spike, **N:** nucleocapsid,
E: envelope, **M:** membrane



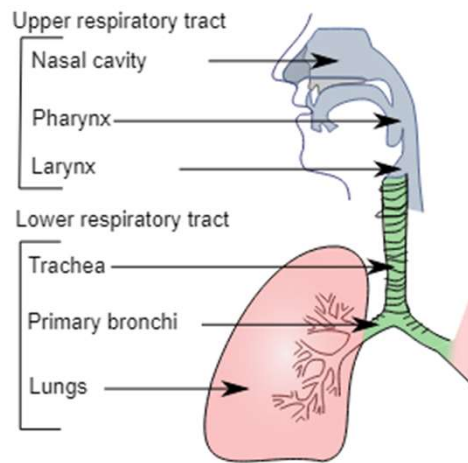
Spike protein binds to the ACE2
(angiotensin-converting enzyme 2)

The cell makes new components of
the virus

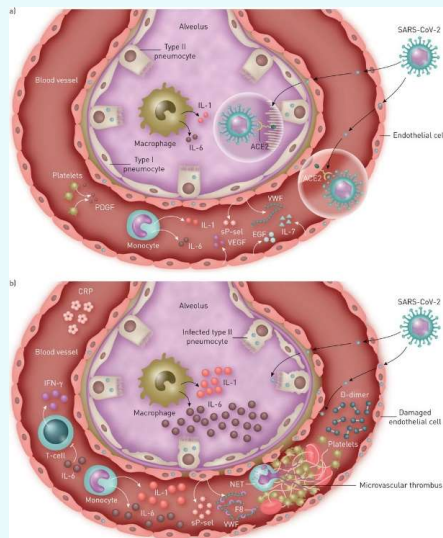


These components are packaged
together to make new COVID-19
virus particles, which are released
to infect more cells

SARS-CoV-2 (COVID-19): pathology



Infects nasal epithelial cells, then travels down respiratory tract to infect lung epithelial cells



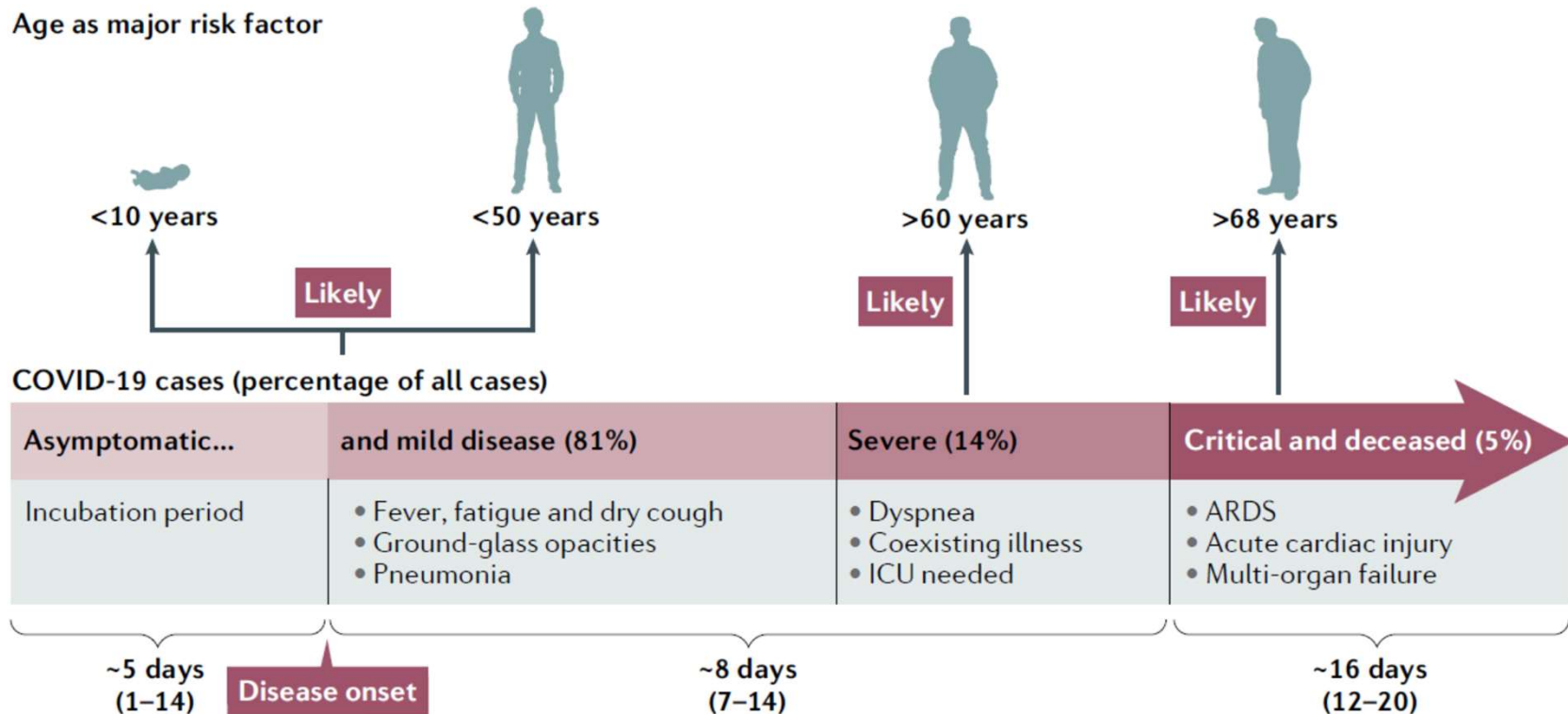
Its rapid proliferation in the lungs can trigger a strong immune response



Cytokine storm syndrome causes acute respiratory distress syndrome and respiratory failure (considered the main cause of death in patients with COVID-19)

SARS-CoV-2 (COVID-19): pathology

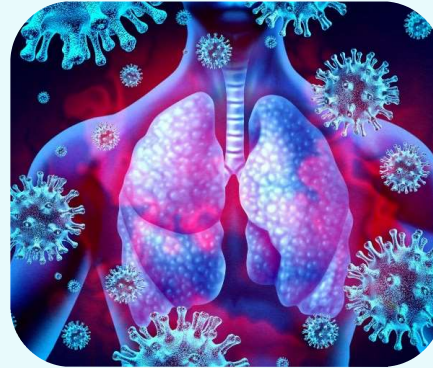
- In general, older men (older 60) with comorbidities and more likely to develop severe respiratory disease that requires hospitalisation
- Younger adults and children more likely to get milder symptoms



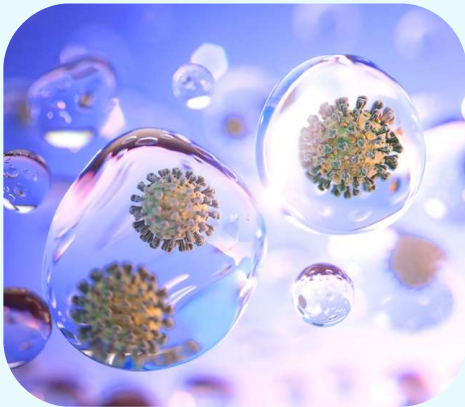
SARS-CoV-2 (COVID-19): transmission



Unlike 'SARS-CoV' – where transmission occurred mainly after illness onset, the risk of COVID-19 transmission is very high at the beginning of infection



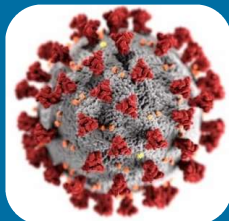
Undocumented infections might account for 79% of documented cases owing to the high transmissibility of COVID-19 during mild disease or the asymptomatic period



Spread in liquid droplets during speech. COVID-19 may also be present in aerosol particles, urine and faeces



Early diagnosis is crucial for preventing the spread of COVID-19



**SARS-CoV-2 infection,
pathology and transmission**



**The clinical trial process and
COVID-19 vaccine approval**

Clinical Trials

Phase I trial

Emphasis on safety. Tests an experimental treatment on a small group of often healthy people (20 to 80) to judge its safety and side effects and to find the correct drug dosage.

Phase II trial

Emphasis on effectiveness. Uses more people (100 to 300). Aims to obtain preliminary data on whether the drug works in people who have a certain disease or condition. Continues to study safety and side effects.

Phase III trial

Number of people range from several hundred to ~3,000 people. Gathers more information about safety and effectiveness, studying different populations and different dosages, using the drug in combination with other drugs.

approval

Phase IV trial

After being approved for use, a drug's effectiveness and safety is monitored in large, diverse populations. Sometimes, the side effects of a drug may not become clear until more people have taken it over a longer period of time.

Temporary Authorisation of COVID-19 vaccine

- Given authorisation for temporary supply by the UK Department of Health and Social Care and the Medicines and Healthcare products Regulatory Agency (MHRA)

No Marketing Authorisation

The MHRA is responsible for issuing Marketing Authorisation. They review and assess the evidence to support a medicine in relation to its marketing and grants a license to sell that medicine.

Temporary Authorisation:

The MHRA has not published specific details about the approval process

Careful and scientific review of the safety, quality and effectiveness data.

Temporary Authorisation : Regulation 174 of Human Medicines Regulations

Only valid for one year

Requires the pharmaceutical companies to complete specific obligations, such as ongoing or new studies
Once comprehensive data on the product as been obtained, standard marketing authorisation can be granted.

It means that only certain batches of vaccine are approved.

Each batch goes through manufacturer's own testing and independent external testing.

Temporary Authorisation of COVID-19 vaccine

Contents

Summary of Product Characteristics

Conditional Marketing Authorisation

Intended for medicinal products that fulfil an unmet medical need

Granted by MHRA where comprehensive clinical data is not yet complete

But is it judged that such data will become available soon

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Regulation 174 of the Human Medicines Regulation

Permits the supply of medicinal products that are given temporary authorisation

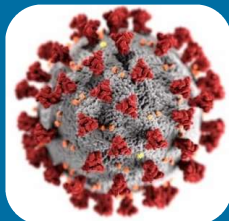
Grants permission for the medicine to be used for active immunization for the prevention of COVID-19

As with any new medicine, the vaccines will be closely monitored to allow quick identification of new safety information

This is a multidose vial which contains 10 doses of 0.5 mL.

One dose (0.5 mL) contains 100 micrograms of messenger RNA (mRNA) (embedded in SM-102 lipid nanoparticles)

AstraZeneca UK Limited



**SARS-CoV-2 infection,
pathology and transmission**



**The clinical trial process and
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COVID-19 vaccines

SARS-CoV-2 (COVID-19) vaccines

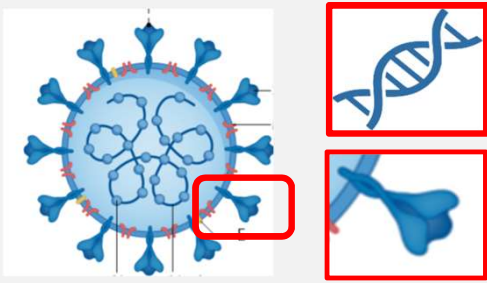
AstraZeneca/Oxford	Non-replicating viral vector (Phase IV clinical trial)	Licensed for active immunization in people aged 18 and over
Pfizer	LNP-encapsulated mRNA (Phase IV clinical trial)	Licensed for active immunization in people aged 12 and over
Moderna	LNP-encapsulated mRNA (Phase IV clinical trial)	Licensed for active immunization in people aged 12 and over

LNP: lipid nanoparticle

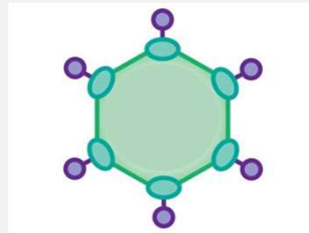
<https://www.medicines.org.uk/emc/product/12333/productinfo> (accessed 2021)

AstraZeneca/Oxford vaccine

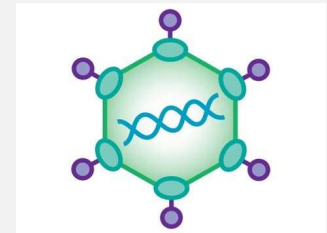
The gene that codes for the spike protein is sequenced and synthesised



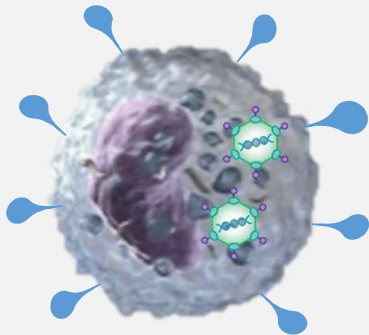
A harmless, weakened adenovirus vector that usually causes the common cold in chimps. It can't grow in humans



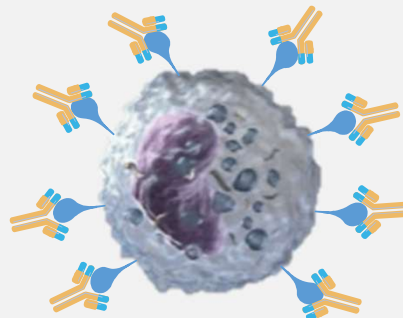
The gene for the COVID-19 spike protein is inserted into the chimpanzee virus



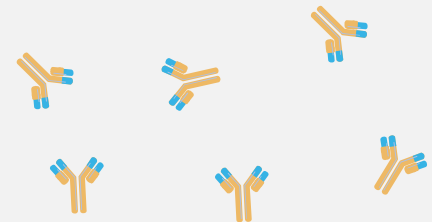
The vector enters human cells which read the spike gene and produce spike protein on their surface



Our immune system is then triggered and produces antibodies to fight of 'the infection'



Our immune system is then primed to attack the coronavirus if it later infects the body



AstraZeneca/Oxford vaccine

Evaluated based on an interim analysis of pooled data from four on-going, randomized, blinded, controlled trials.

Phase I/II Study

UK trial on healthy adults aged 18 to 55

Phase II/III Study

UK trial on adults aged 18 or over (including elderly)

Phase III Study

In adults aged 18 or over (including elderly) in Brazil

Phase I/II Study

In adults aged 18 to 65 in South Africa

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graph TD; A[Phase I/II Study] --> D[Licensed seasonal flu and pneumococcal vaccinations permitted ≥ 7 days before or after]; B[Phase II/III Study] --> D;
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Licensed seasonal flu and pneumococcal vaccinations permitted ≥ 7 days before or after

Excluded from study:

- Patients with a history of anaphylaxis or angioedema
- Patients with severe and/or uncontrolled cardiovascular, gastrointestinal, liver, renal, endocrine/metabolic disease and neurological illnesses
- Immunosuppressed patients

AstraZeneca/Oxford vaccine

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Phase III Study

In adults aged 18 or over (including elderly) in Brazil

Phase I/II Study

In adults aged 18 to 65 in South Africa

Exceeded threshold of at least 5 confirmed COVID-19 cases per study

5,807 = vaccine

2 doses of vaccine / control

COVID cases = 0.5%

5,829 = control

(dosing interval between 4 to 26 weeks)

COVID cases = 1.7%

= 70.4% vaccine efficacy

AstraZeneca/Oxford vaccine

Evaluated based on an interim analysis of pooled data from four on-going, randomized, blinded, controlled trials.

Phase I/II Study

UK trial on healthy adults
aged 18 to 55

Phase II/III Study

UK trial on adults aged 18
or over (including elderly)

Phase III Study

In adults aged 18 or over
(including elderly) in Brazil

Phase I/II Study

In adults aged 18 to 65 in
South Africa

Analysis of results from all four studies

8,597 = vaccine

2 doses of vaccine /
control

COVID cases = 1%

8,581 = control

(dosing interval between
4 to 26 weeks)

COVID cases = 2.9%

= 66.7% vaccine efficacy

AstraZeneca/Oxford vaccine

Given authorisation for temporary supply which grants permission to be used for active immunisation in people 18 years and over

As with any new medicine, it will be closely monitored. Healthcare professionals are advised to report any suspected side effects.

SARS-CoV-2 (COVID-19) vaccines

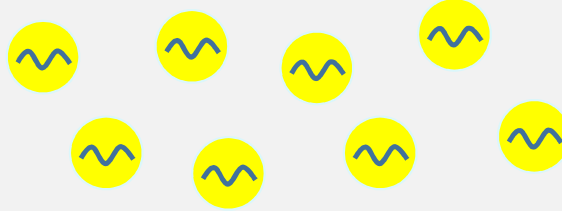
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Pfizer vaccine

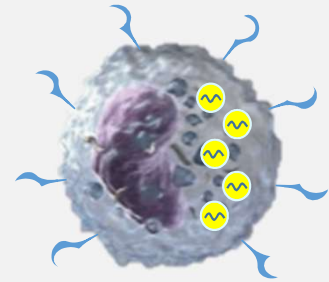
Single strand of messenger RNA (mRNA), produced by cell-free *in vitro* transcription that codes for the COVID spike protein



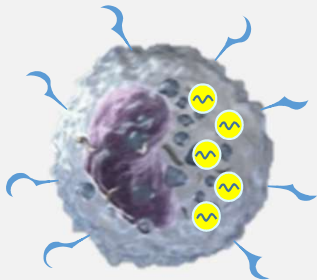
The mRNA is formulated in lipid nanoparticles, which deliver it into our cells of the body



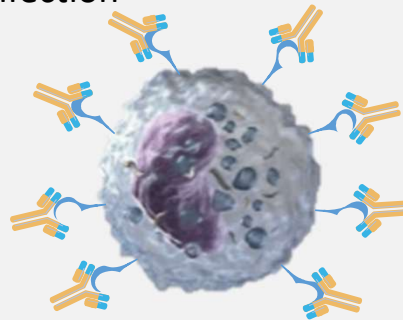
Once inside, the mRNA instructs our cells to make a harmless piece of the COVID-19 spike protein



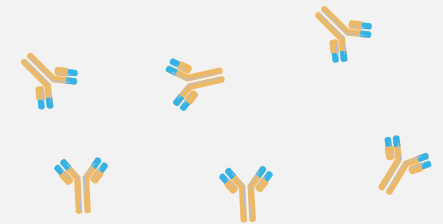
The mRNA is destroyed. The immune system recognises the spike protein piece doesn't belong there



Our immune system is then triggered and produces antibodies to fight of 'the infection'



Our bodies have learned how to protect against future infection from the COVID-19 virus



Pfizer vaccine

Vaccine efficacy evaluated in two clinical studies conducted in the US, Europe, Turkey, South Africa and South America

Study 1

60 participants, aged 18 - 55

Study 2

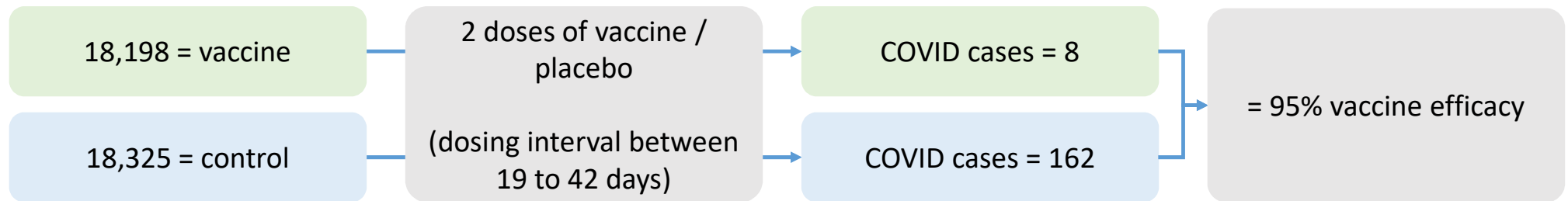
Multi-centre, placebo-controlled efficacy study.
44,000 participants aged 12 year and over.

Licensed seasonal flu and pneumococcal vaccinations permitted ≥ 14 days before or after

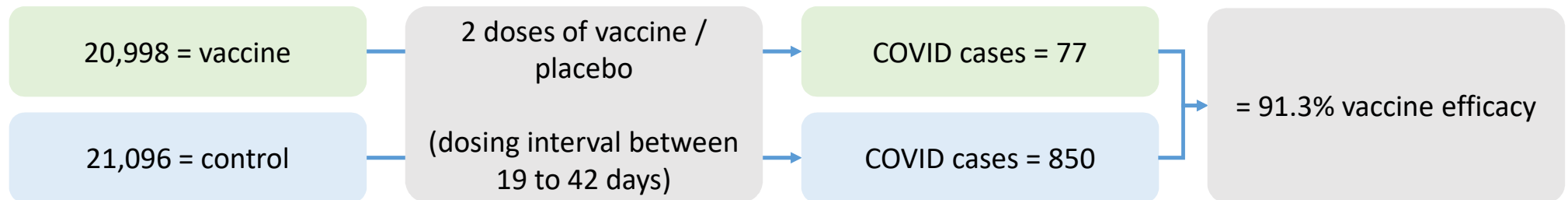
Excluded from study:

- Immunosuppressed patients
- Patients with a previous diagnosis of COVID-19 disease

Pfizer vaccine: Study 2 details



Updated efficacy analyses: representing up to 6 months after second dose



Given authorisation for temporary supply which grants permission to be used for active immunisation in people aged 12 and over

As with any new medicine, it will be closely monitored. Healthcare professionals are advised to report any suspected side effects.

SARS-CoV-2 (COVID-19) vaccines

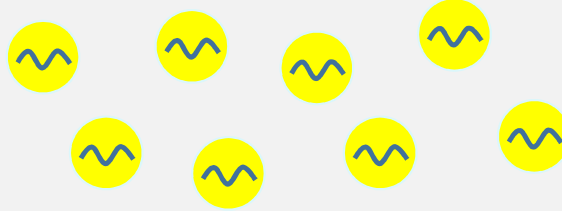
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Moderna	LNP-encapsulated mRNA (Phase IV clinical trial)	Licensed for active immunization in people aged 12 and over

Moderna vaccine

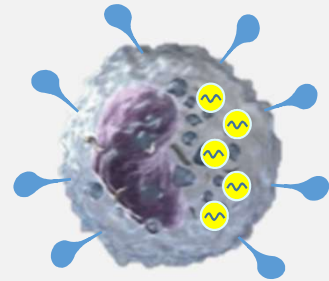
Single strand of messenger RNA (mRNA), produced by cell-free *in vitro* transcription



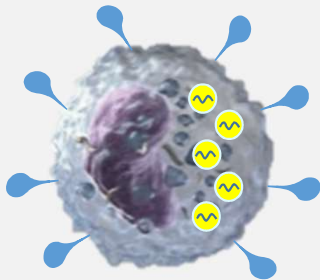
The mRNA is formulated in lipid nanoparticles, which deliver it into our cells of the body



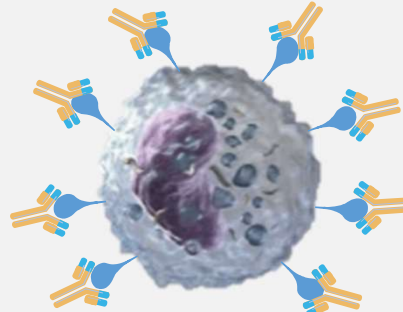
Once inside, the mRNA instructs our cells to make a **the complete** COVID-19 spike protein



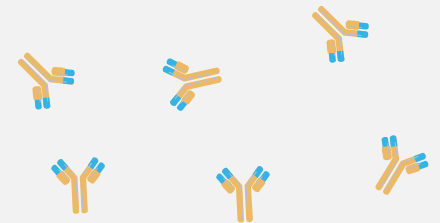
The mRNA is destroyed. The immune system recognises the spike protein doesn't belong there



Our immune system is then triggered and produces antibodies to fight of 'the infection'



Our bodies have learned how to protect against future infection from the COVID-19 virus



Moderna vaccine

Vaccine efficacy evaluated in a randomised, placebo controlled Phase 3 trial

Licensed seasonal flu and pneumococcal vaccinations permitted ≥ 14 days before or after

Excluded from study:

- Immunosuppressed or people who had received immunosuppressants within 6 months
- Participants that were pregnant
- People with a known history of COVID-19 infection

14,134 = vaccine

2 doses of vaccine / placebo

COVID cases = 11

14,073 = control

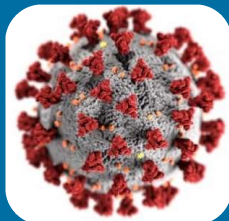
(dosing interval between 25 to 352 days)

COVID cases = 185

= 94.1% vaccine efficacy

Granted **conditional approval**: this means that further evidence is awaited.

As with any new medicine, it will be closely monitored. Healthcare professionals are advised to report any suspected side effects.



**SARS-CoV-2 infection,
pathology and transmission**



**The clinical trial process and
COVID-19 vaccine approval**



COVID-19 vaccines



**Child vaccination and booster
vaccinations**

COVID-19: vaccination in children

Factors to consider in relation to
COVID-19 vaccination
of children

Figure 1 Summary of benefits and risks of vaccinating children against COVID-19. PIMS-TS, paediatric inflammatory multisystem syndrome-temporally associated with SARS-CoV-2.

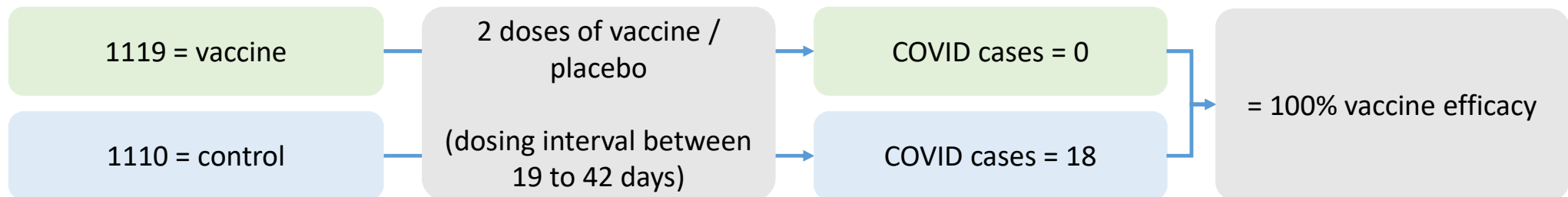
Zimmerman *et al.*, (2021)

COVID-19: vaccination in children

- Both Pfizer and Moderna vaccine granted conditional approval for vaccination of 12 – 18 year olds
- People aged 16 to 17 vaccinated from August 2021
- People aged 12 to 15 vaccinated from September 2021



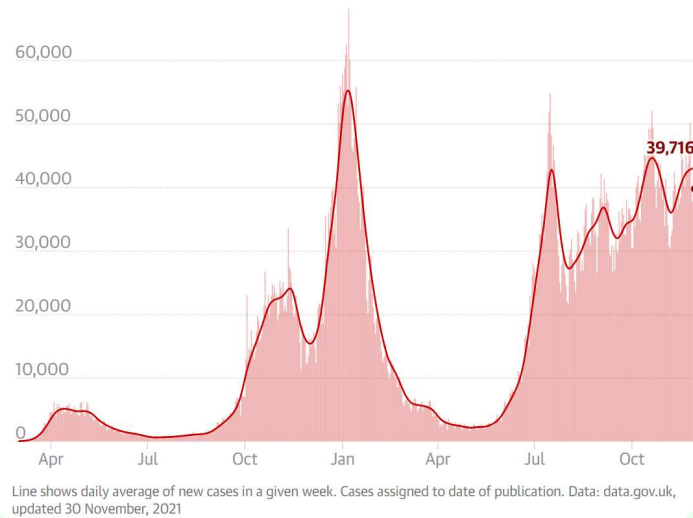
Pfizer vaccine efficacy in adolescents 12 to 15 years old: with or without prior COVID infection



<https://www.gov.uk/government/news/all-young-people-aged-16-and-17-in-england-to-be-offered-vaccine-by-next-week>
<https://www.cdc.gov/coronavirus/2019-ncov/vaccines/different-vaccines/mrna.html> (Accessed November 2021)

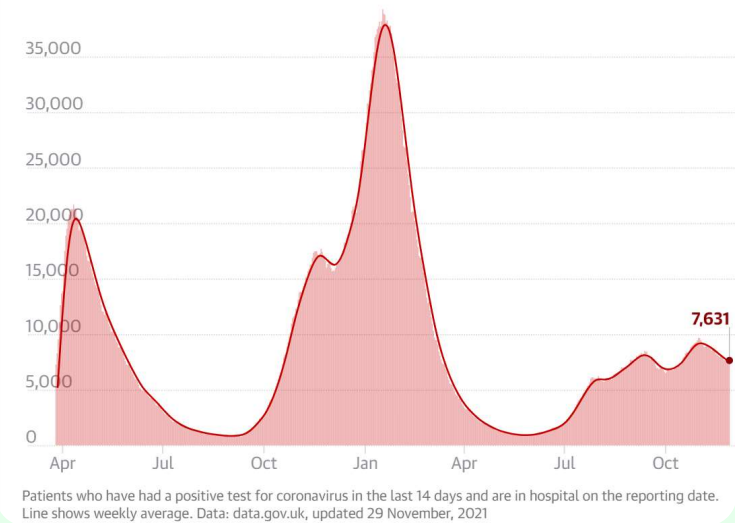
COVID-19: booster vaccines

UK: new coronavirus cases per day



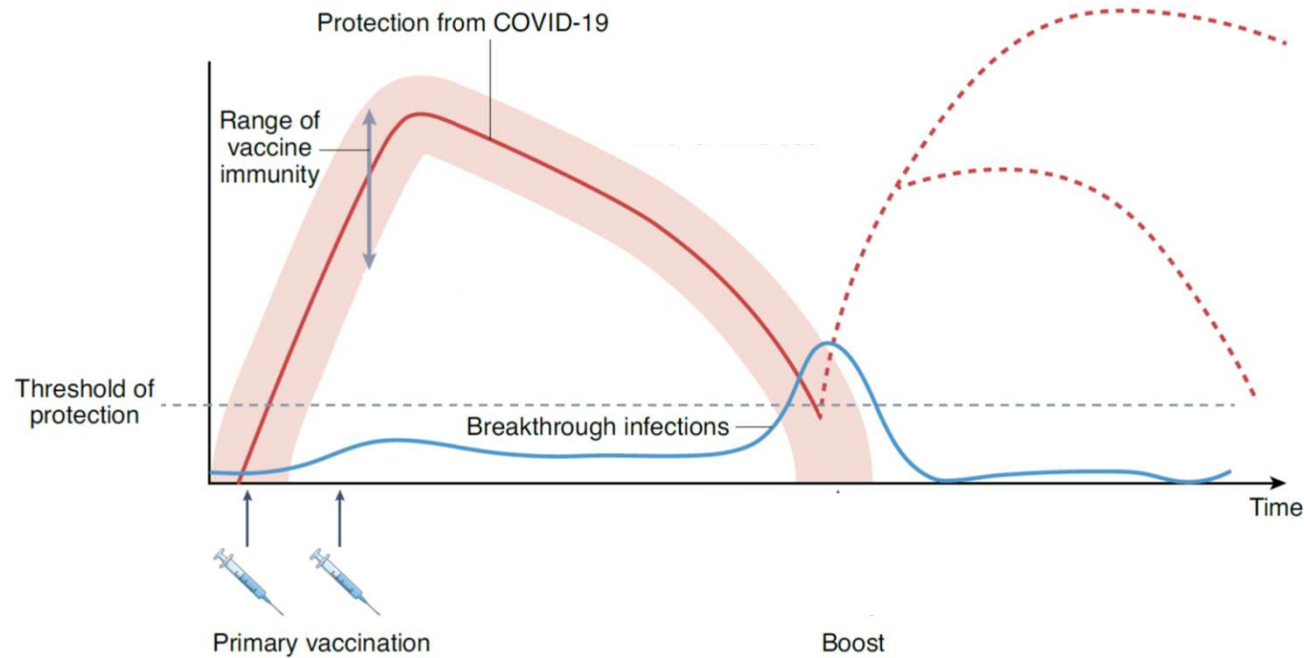
- Infection post vaccination may provide a natural antibody boost
- Increasing signs that population-level immunity might be waning. Breakthrough infections are on the rise
- Needs to be monitored closely

UK: people in hospital with coronavirus each day



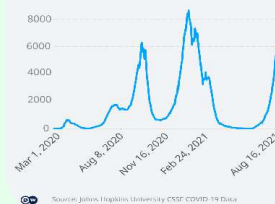
- Whilst protection against hospitalisation and death is maintained, 'booster' vaccinations may not be needed
- Need to increase vaccine effectiveness in vulnerable populations (immunosuppressed, elderly)
- Need to maintain protection against emerging variants of concern with immune evasion potential.

COVID-19: booster vaccines



Studies in Israel showed for people vaccinated at least 5 months earlier, a 3rd dose increased immunity 5-7 fold.

COVID-19 daily new cases in Israel

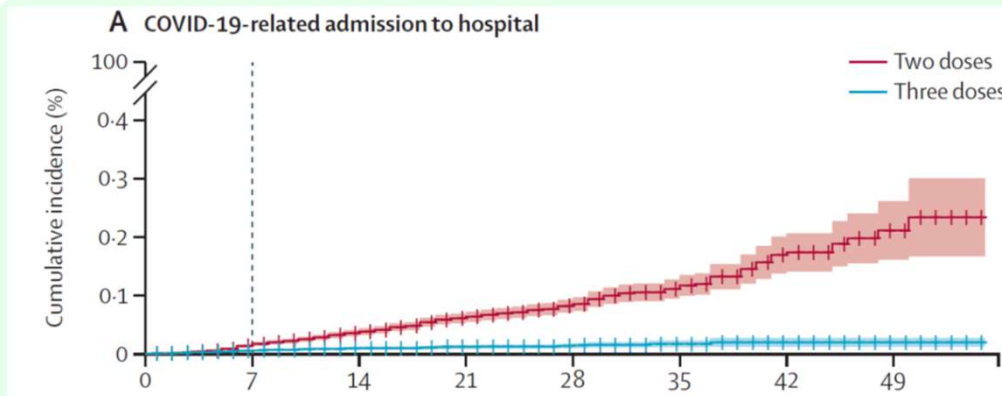


This was matched by a reduction (11.3 fold) in breakthrough infections.

COVID-19: booster vaccines



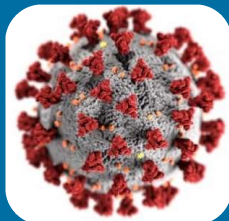
The Pfizer OR Moderna vaccine may be given as a booster after second dose of either an mRNA or adenovirus-vectored COVID-19 vaccine when the potential benefits outweigh any potential risks.



The result of a trial on a booster Pfizer vaccine showed vaccine effectiveness of **93%** for admission to hospital—during a time when Delta was the prevalent strain

- 728,321 people (median age of 52 years) given a third dose of the Pfizer vaccine in Israel
- Vaccine effectiveness of **92% for severe disease**
- Vaccine effectiveness of **81% for COVID-19-related death**

‘A third dose of mRNA vaccine is effective in reducing severe COVID-19 related outcomes for patients who have received two doses at least 5 months before.’



**SARS-CoV-2 infection,
pathology and transmission**



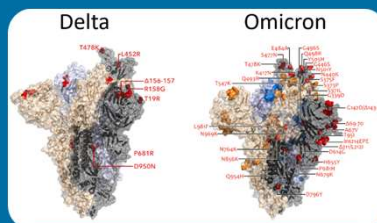
**The clinical trial process and
COVID-19 vaccine approval**



COVID-19 vaccines



**Child vaccination and booster
vaccinations**

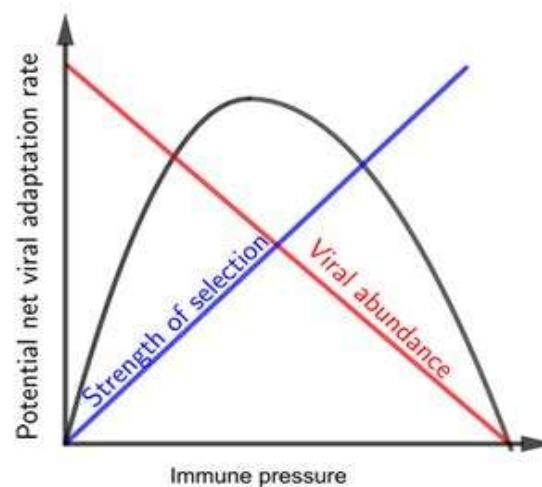


COVID-19 variants

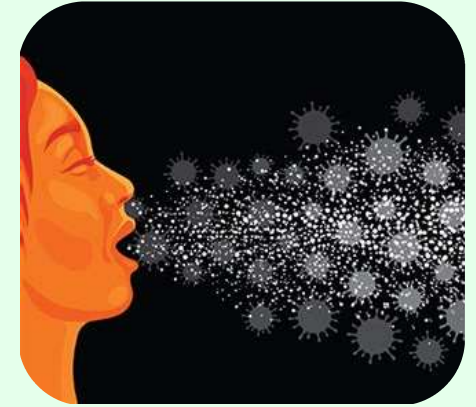
COVID-19: variants



Evidence suggests that vaccine efficacy against severe disease may be preserved against current identified VOC

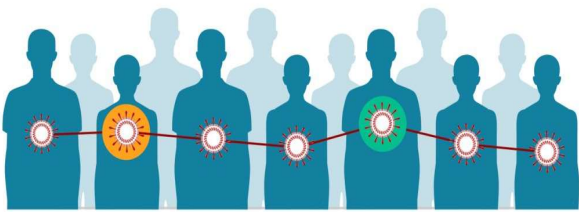


Risk of '**vaccine escape**' : vaccine generated immunity is insufficient to provide protection against disease

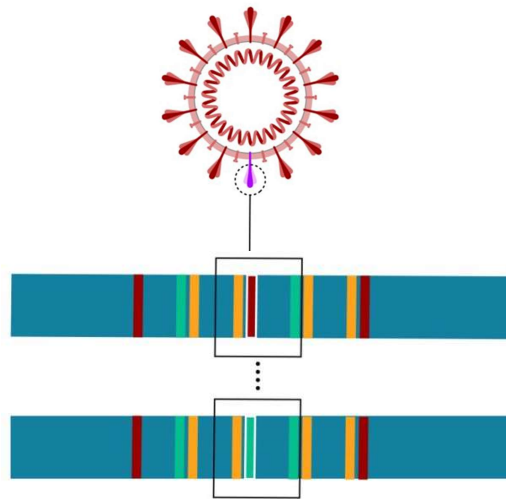


High virus transmission and immunity generated by vaccination / previous infection may drive selection of 'escape mutants'

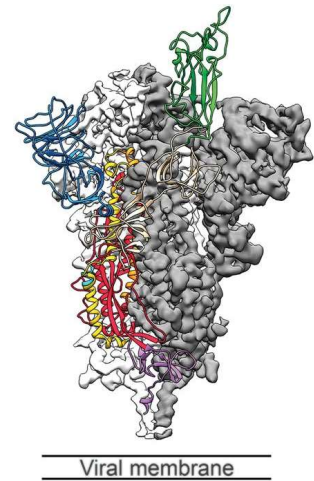
COVID-19: variant surveillance



All first-generation spike-based vaccines in clinical use are generated from the ancestral Wuhan *spike* gene sequence



This raises the potential for loss of vaccine efficacy as SARS-CoV-2 accumulated mutations during viral evolution



Mutations in the gene sequence that codes for the RBD of the spike protein are particularly worrying because this is the part anti-COVID-19 antibodies recognise

RBD: receptor binding domain

Clemens *et al.*, (2021) <https://www.bbc.co.uk/news/health-59418127> (Accessed November 2021)

COVID-19 BETA variant

- First identified in South Africa in late 2020
- People infected are more likely to need critical care and to die than those infected with other variants
- Seems to be more resistant to immunity generated by vaccines and previous infections than other variants
- Beta variant contains 10 changes across the spike protein



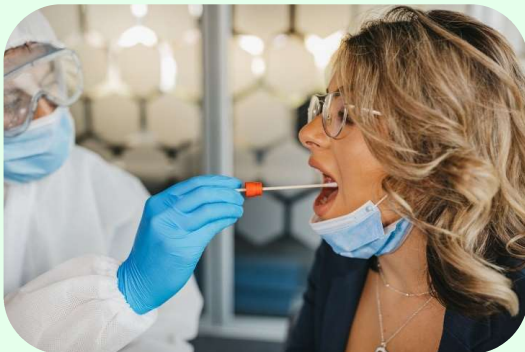
- AstraZeneca/Oxford began phase II/III trial to test new COVID-19 vaccine against the Beta variant in June 2021
- Will recruit ~2,250 participants across UK, South Africa, Brazil and Poland
- Vaccine (AZD2816) designed using same adenoviral vector platform as the first AstraZeneca vaccine
- With minor genetic alterations to the spike protein based on the Beta variant

<https://www.nature.com/articles/d41586-021-02177-3> <https://www.astrazeneca.com/media-centre/press-releases/2021/first-covid-19-variant-vaccine-azd2816-phase-ii-iii-trial-participants-vaccinated.html>

(Accessed November 2021)

COVID-19 DELTA variant

- A more transmissible COVID-19 variant
- Caused sharp rises in infections in many countries, including those with relatively high vaccine coverage (e.g. UK)
- Linked to heightened severity
- The Delta quickly became the dominant variant after being classified as a 'variant of concern' in April 2021



- Large community surveillance study in UK
- Found reduced vaccine efficiency (both Pfizer and AstraZeneca vaccine)
- Delta variant infection resulted in similar viral burden to those infected without vaccination
- Similar percentages of infected people reporting symptoms
- **Both Pfizer and AstraZeneca vaccines remain protective against the Delta variant**

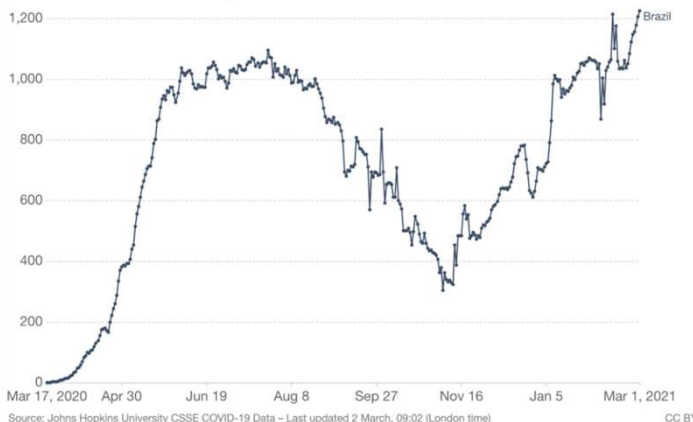
COVID-19 ZETA and GAMMA variant

- Zeta and Gamma variants emerged in Brazil towards the end of 2020
- Infection with these variants has been described in adults that already have had COVID-19 and the alpha variant
- Both variants have mutations in the virus spike protein (1 mutation in Zeta, 3 mutations in Gamma)



Daily new confirmed COVID-19 deaths

Shown is the rolling 7-day average. Limited testing and challenges in the attribution of the cause of death means that the number of confirmed deaths may not be an accurate count of the true number of deaths from COVID-19.

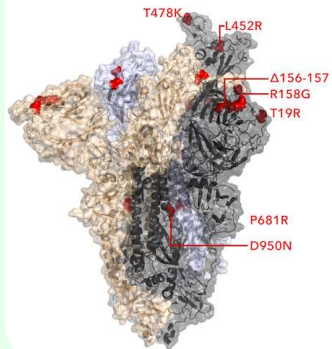


- Post-hoc exploratory analysis of a Phase III trial to investigate the efficiency of the existing AstraZeneca vaccine against the Zeta and Gamma variants in Brazil
- The vaccine provided protection against severe and death
- Vaccine efficiency after 2 doses of vaccine was 69% against the Zeta variant
- There were relatively few cases of the Gamma variant – protection from symptomatic disease may be reduced in comparison with Zeta (64%)

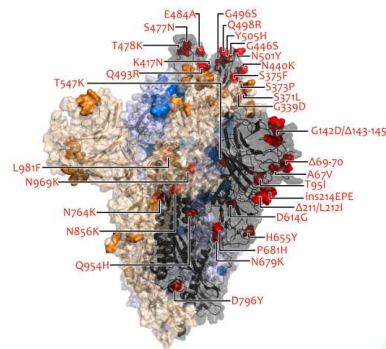
<https://www.americasquarterly.org/article/brazils-handling-of-covid-19-is-a-global-emergency/> (Accessed November 2021) Clemens *et al.*, (2021)

COVID-19: OMICRON

Delta



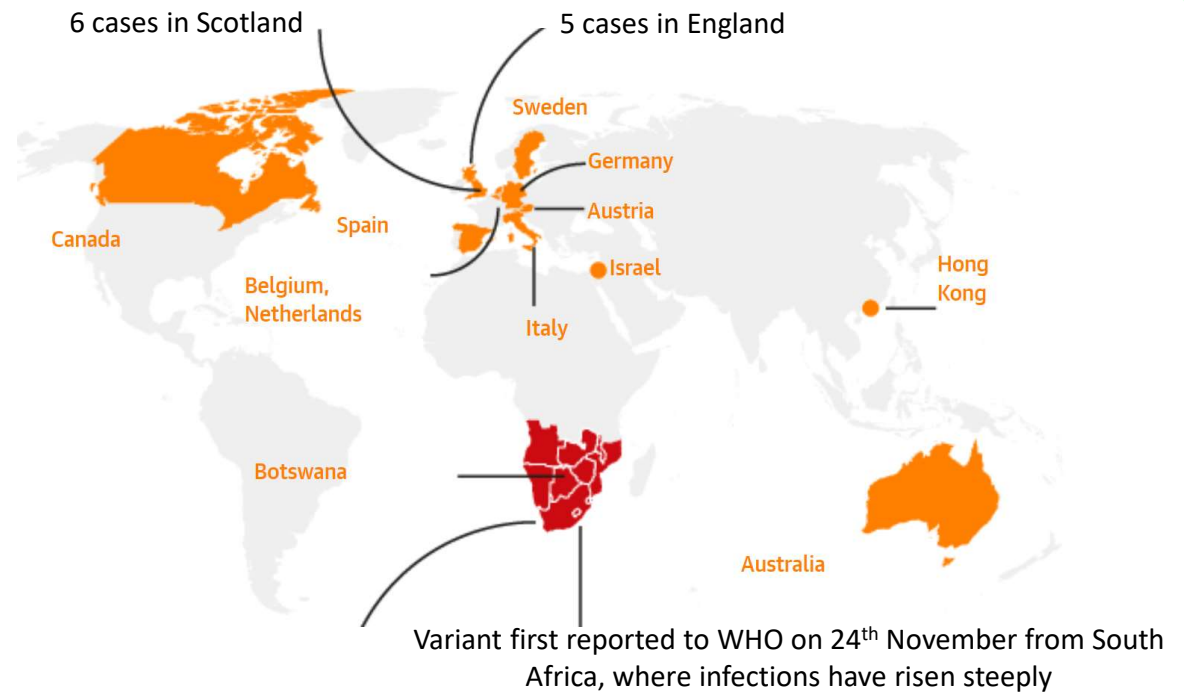
Omicron



A highly mutated variant: including more than 30 mutations in the spike protein.

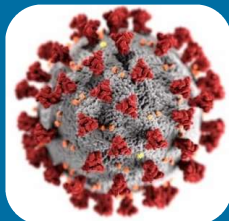
It is unknown whether these will enable OMICRON to escape immunity induced by vaccines and previous infections.

Unknown whether it causes more severe disease.



Vaccine manufacturers are looking into OMICRON to see how vaccines work against it, and also are rapidly advancing updated / OMICRON-specific boosters.

<https://www.sciencefocus.com/news/omicron-covid-variant/>
<https://www.theguardian.com/world/2021/nov/29/omicron-covid-variant-poses-very-high-global-risk-says-who>
<https://gothamist.com/news/why-it-will-take-two-weeks-to-learn-if-omicron-impacts-covid-19-vaccines> (Accessed November 2021)



**SARS-CoV-2 infection,
pathology and transmission**



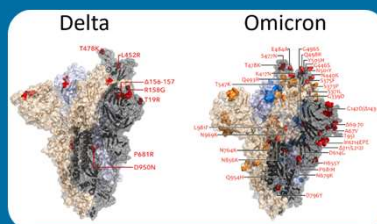
**The clinical trial process and
COVID-19 vaccine approval**



COVID-19 vaccines



**Child vaccination and booster
vaccinations**



COVID-19 variants

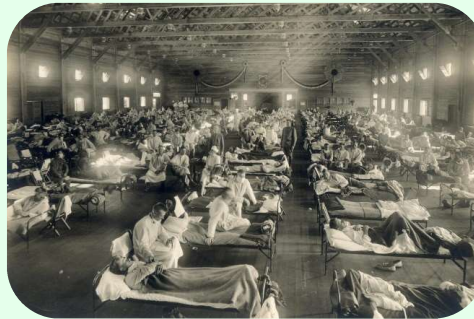


COVID-19 and the future

Comparison with the influenza vaccine



Flu vaccines are 'leaky'; effectiveness is never higher than 50 – 60% and some years much lower



The 'flu jab' aims to reduce outbreaks and prevent severe complications. Preventing mild disease is less critical.



COVID vaccines are more efficient than flu vaccines



Flu vaccines are updated globally twice a year on the basis of recommendations from the WHO consultation

Virus selection for annual flu jabs

Disease and virus surveillance

Virus isolation and antigenic characterisation

Genetic characterisation of viruses

Human serology studies with influenza vaccine viruses

Virus fitness forecasting

Vaccine effectiveness

Selection of candidate vaccine viruses

Lessons learnt about vaccine development

Following a traditional research and development pipeline, it takes between 5 – 10 years to develop a vaccine for a new infectious agent

For SARS-CoV2, it took 300 days.



The faster a vaccine is deployed, the faster an outbreak can be controlled

Rapid response platforms could move from sequence to clinical trials in a matter of weeks rather than months or even years – such as mRNA and DNA tech



Surveillance for variants also essential : mRNA platforms will be able to generate new vaccines very rapidly

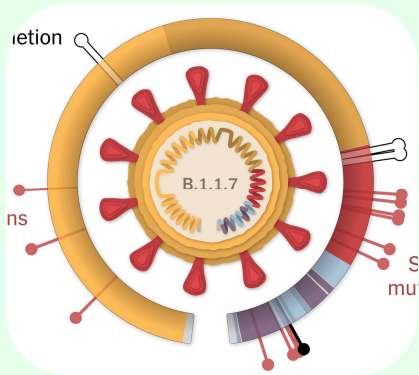


Ongoing pharmacovigilance for vaccines under development and post approval with risk management plans in place



Seasonal flu vaccines typically don't require new trials, but new trials will be necessary for new COVID vaccines.

The future



With the evolution of new variants and waning immunity over time, vaccinated people may get infected, albeit with milder symptoms.

Even natural infection provides only short term protection.



Herd immunity doesn't work well when a virus is as transmissible as COVID-19 is.

It seems probable that COVID-19 will not be eliminated from the human population.



Inevitable that re-vaccination will be necessary to counter waning immunity and against new variants.

Multivalent vaccines are also in development.



Development of better therapeutics and anti-viral agents not affected by changes in variants for people that develop severe symptoms despite vaccination.

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